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described is applicable to many other III-V compound semiconductors and has the potential to be modified for obtaining similar improvements in ohmic contacts to *n*-type GaAs and related compounds.

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Drive-level capacitance profiling: Its application to determining gap state densities in hydrogenated amorphous silicon films

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We have developed a new method of capacitance profiling which is greatly superior to standard profiling techniques for materials like amorphous silicon that contain high densities of deep gap states. This technique, which makes use of the variation of the junction capacitance with the alternating voltage amplitude, is largely immune to the effects of surface states and is readily interpreted in terms of a simple integral over the density of states, $g(E)$. This method is applicable to both doped and semi-insulating undoped samples.

Among the most useful methods used to characterize semiconductors are capacitance-voltage (*C-V*) profiling measurements. Methods such as Miller¹ and Copeland² profiling give the net doping concentration versus distance in the vicinity of a barrier interface formed on the material under study. In amorphous silicon, however, or indeed any semiconductor with a large concentration of deep gap states, such techniques are not so easily interpreted because the depletion charge density is not fixed but depends on the applied bias. Also, deep states can give rise to an enormous spectrum of detrapping times so such measurements become very dependent on the time scale of the measurement process.

In this letter we describe an alternative capacitance profiling procedure by which the spatial variation of the gap state concentration in amorphous silicon can be easily determined. The analysis required to yield gap state densities is extremely straightforward and this technique is greatly immune to anomalies near barrier interfaces which significantly influence many of the other kinds of measurements.^{3,4} Finally, unlike deep level transient spectroscopy⁵ (DLTS) and related techniques which require doped samples with high conductivities, this method is readily applicable to the study of undoped films and may be used to map out $g(E)$ in such materials over a limited range of the mobility gap.

Our technique makes use of the dependence of the junction capacitance, C , on the amplitude of the applied alternating voltage. In the limit of vanishingly small amplitude the

junction capacitance C_0 is given by^{6,7}

$$C_0 = \epsilon A \rho_e / (\epsilon F_e + x_e \rho_e), \quad (1)$$

where A is the junction area, ϵ is the dielectric constant, and x_e denotes the distance from the barrier interface where the Fermi level E_F lies at an energy E_e from the nearest mobility edge E_c such that states near E_F will be emitted (and recaptured) on a time scale

$$1/\omega = \nu^{-1} \exp(E_e/kT). \quad (2)$$

Here ω is the (angular) frequency of the alternating voltage and ν is the usual prefactor for thermal emission and lies in the range 10^{12} – 10^{13} s⁻¹.⁵ At locations further from the barrier than x_e gap states at E_F will be shallower than E_e and thus respond much faster than $1/\omega$; at locations closer than x_e states at E_F will respond exponentially more slowly and will be essentially frozen with respect to the oscillating component of the applied voltage.

The quantities F_e and ρ_e in Eq. (1) refer to the value of the electric field and charge density at the point x_e , respectively. The value ρ_e is given simply by the integral over the density of states:

$$\rho_e = \rho(E_e) = q \int_{E_c - E_e}^{E_F^0} g(E) dE. \quad (3)$$

Here E_F^0 refers to the position of Fermi level in the (neutral) bulk material and q is the electronic charge. Note that Eq. (1) is valid only if the film properties are slowly varying beyond

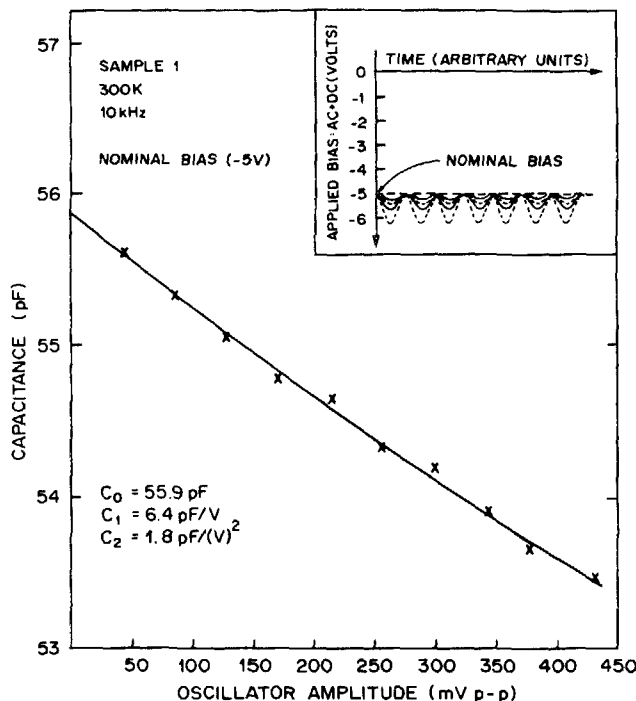


FIG. 1. Variation of junction capacitance with the p - p alternating voltage drive level. The measured points are fit to a second degree polynomial to obtain the coefficients given. The inset shows the manner in which the dc bias and alternating voltage amplitude are varied to maintain a constant low magnitude extremal value for each nominal dc bias level.

the point x_e . However, the spatial variation of the material in the region closer than x_e does *not* affect the validity of this result; indeed, sample attributes such as surface or near interface anomalies lying closer to the barrier than x_e do *not* change either ρ_e or F_e and only modify the junction capacitance by the fact that they help determine the overall position of x_e .

If the p - p alternating voltage drive level δv is not zero we obtain higher order corrections to the junction capacitance. Specifically

$$C = C_0 + C_1 \delta V + C_2 (\delta V)^2 + \dots, \quad (4)$$

where C_0 is given by Eq. (1) and

$$C_1 = -\frac{1}{2} \epsilon A \rho_e^2 / (\epsilon F_e + x_e \rho_e)^3. \quad (5)$$

We utilize this variation with drive level amplitude to define the drive-level profiling technique which plots the quantity

$$N_{DL} = -C_0^3 / 2q\epsilon A^2 C_1 = \rho_e / q \quad (6)$$

versus the spatial quantity $W = \epsilon A / C_0$. The advantages of this technique are obvious from the above discussion. (1) The quantity N_{DL} is independent of x_e and therefore also independent of surface states or near interface anomalies in the film. (2) The charge density N_{DL} is directly and simply related to the density of states in the material, $g(E)$, through Eq. (3). (3) The portion of the density of states to be profiled determined by the quantity E_e may be experimentally selected by changing T and/or ω . Note that while the quantity $\epsilon A / C_0 = x_e + (\epsilon F_e / \rho_e)$ does not give the actual location at which ρ_e is measured, F_e / ρ_e is roughly constant at a given temperature so that variations in W correspond to actual variations over approximately the same distance scale.

At this point two important experimental details should

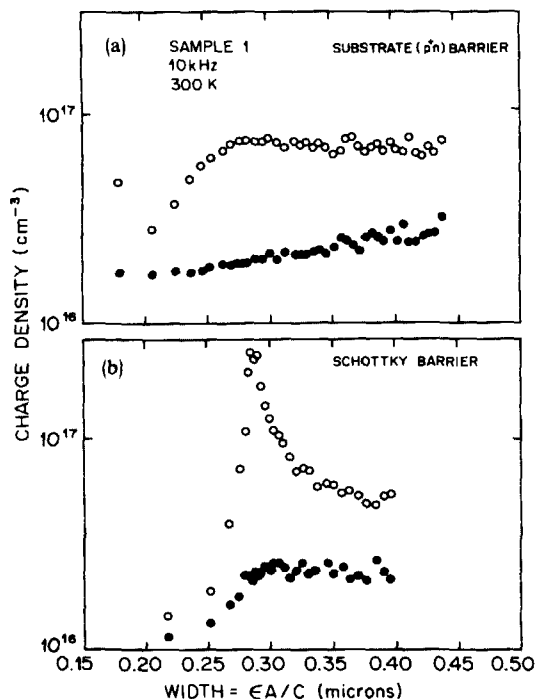


FIG. 2. Variation of N_{DL} (filled circles) and N_{CV} (open circles) with $W = \epsilon A / C_0$ for (a) the reverse bias at the p^+ - n substrate junction and (b) the reverse bias applied to the front Pd Schottky barrier junction.

be mentioned. First, the *nominal* dc bias level is taken to be the low magnitude extremal value of the oscillator voltage *plus* dc bias, so that the oscillator amplitude and dc value must be changed in tandem to keep this extremal value constant (see inset, Fig. 1). This procedure is followed to correspond to the assumptions under which Eq. (5) was derived. Second, in profiling films at lower temperatures, we always started with the highest magnitude bias which was applied at moderate temperatures prior to cooling. This procedure ensured that the space-charge region was essentially in an equilibrium situation throughout the profiling measurement.

To demonstrate the drive-level profiling technique we chose two rf glow discharge a -Si:H samples. Sample 1 was n -type doped (PH_3 at 100 vppm) and sample 2 was undoped. Both samples were grown with a rf power density of 85 mW/cm² and a substrate temperature of 250 °C on p^+ -doped crystalline silicon substrates. Palladium Schottky barriers were deposited on the film surface. This provided a barrier region at both the top and bottom interfaces of each sample and allowed either part of the film to be profiled by the choice of the dc bias polarity.

Figure 1 shows the variation of the 10-kHz junction capacitance with alternating voltage amplitude for sample 1 at room temperature. For each set of measurements we determined both the intercept C_0 as well as the limiting slope C_1 by fitting these points to a second degree polynomial. Repeating such measurements at different bias gave the drive-level profiles displayed in Fig. 2. In this figure we display the profiles for both the p^+ and Schottky barrier interfaces together with the standard C - V profiles determined by the variation of C_0 with bias.¹ An impressive demonstration of the immunity of N_{DL} to surface states comes by comparing the profiles on the two interfaces. The presence of the marked peak for N_{CV} at the front barrier was found to de-

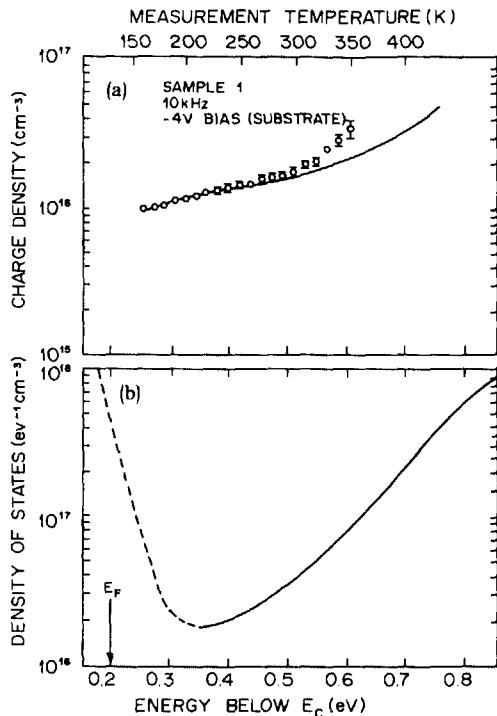


FIG. 3. (a) Variation of N_{DL} for sample 1 at -4 V bias as a function of the measurement temperature as given by the top scale with the corresponding value of E_c indicated below. The error bars indicate the range of values for N_{DL} between -3.5 and -4.5 V applied bias. The solid line is derived by integrating the density of states shown in (b). (b) Density of states for sample 1 derived from DLTS measurements (solid line). The dotted line portion is inferred from thermally stimulated capacitance measurements.

pend on the sample surface treatment before Pd deposition. However, N_{DL} shows no such anomaly and agrees quite well with the rear barrier results.

In Fig. 3(a) we display the variation of the 10-kHz drive-level capacitance versus temperature (at fixed bias) corresponding to a range of energy for E_c from roughly 0.3 to 0.6 eV. The solid line in Fig. 3(a) is the integral of the DLTS density of states determined for this sample shown in Fig. 3(b). Agreement with the N_{DL} data, as predicted by Eq. (3), is seen to be quite good for energies shallower than $E_c - 0.5$ eV. The difference between the two curves at deeper energies is likely due to sample area spreading⁸ which would produce the type of deviation observed. Alternatively, it might be due to a variation of the exponential prefactor for two such significantly different times scales (100 ms for DLTS vs 15 μ s for N_{DL}).

Figure 4(a) shows the results of a 100-Hz drive-level density versus temperature for sample 2 (our undoped sample). In Fig. 4(b) we show a Gaussian-shaped density of states characterized by just three parameters as indicated which fits these data. These results represent the most accurate determination of $g(E)$ for an undoped sample from junction capacitance measurements. One should recognize the similarity of this $g(E)$ to that previously determined for n -type

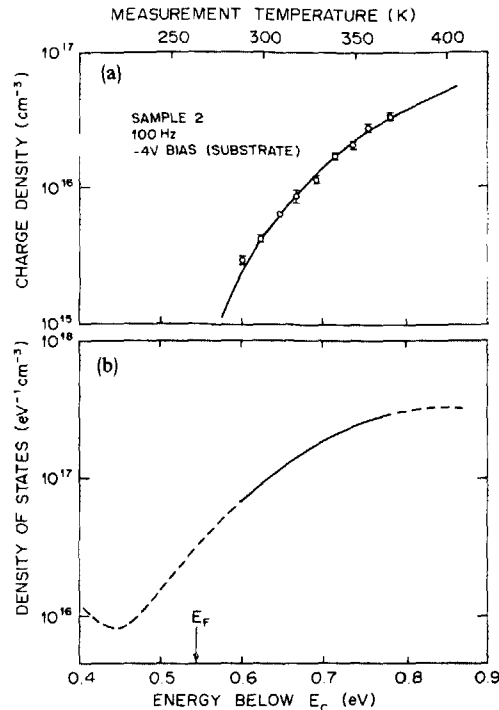


FIG. 4. (a) Variation of N_{DL} for sample 2 at -4 V bias as a function of the measurement temperature with error bars as in Fig. 3(a). The solid line is derived by integrating the density of states shown in (b). (b) Result of Gaussian-shaped density of states fit to data points of (a): Gaussian peak position at $E_c - 0.85$ eV, width parameter of 140 meV, and magnitude (at peak) of 3.3×10^{17} eV⁻¹ cm⁻³. The minimum at $E_c - 0.45$ eV occurs if we assume an exponential band tail with a 35-meV characteristic energy.

doped α -Si:H samples.⁵ Other undoped α -Si:H samples recently analyzed in this manner⁹ give densities of states quite similar to that shown in Fig. 4(b) but with up to an order of magnitude lower defect densities.

In conclusion, the drive-level capacitance profiling technique offers many important advantages over traditional profiling techniques for semiconductors in which a substantial fraction of the depletion region charge is due to deep gap states. The technique can also be used to determine the density of states in both doped and undoped samples.

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